

Extrema and the Mean Value Theorem

ANSWER KEY



Critical points and extrema

- 1 Find the critical points of $f(x) = 2x^3 - 3x^2 - 12x + 5$.
 $f'(x) = 6x^2 - 6x - 12 = 6(x^2 - x - 2) = 6(x - 2)(x + 1)$. Critical points: $x = 2$ and $x = -1$.
- 2 Find the absolute maximum and minimum of $f(x) = x^3 - 3x$ on $[-2, 3]$, checking critical points and endpoints.
 $f'(x) = 3x^2 - 3 = 0$ gives $x = \pm 1$ (both in $[-2, 3]$). Values: $f(-2) = -2$, $f(-1) = 2$, $f(1) = -2$, $f(3) = 18$. Absolute max: $f(3) = 18$. Absolute min: $f(-2) = f(1) = -2$ (tied).

The Extreme Value Theorem

- 3 State the Extreme Value Theorem and explain why it guarantees $f(x) = x^2 - 4x$ has an absolute max and min on $[0, 5]$.
The EVT states that a continuous function on a closed interval $[a, b]$ attains both an absolute maximum and an absolute minimum on that interval. Since $f(x) = x^2 - 4x$ is continuous (a polynomial) and $[0, 5]$ is closed, the EVT guarantees both extrema exist.

Rolle's Theorem

- 4 Verify that Rolle's Theorem applies to $f(x) = x^2 - 4x + 3$ on $[1, 3]$, and find the value(s) of c guaranteed by the theorem.
 $f(1) = 0$ and $f(3) = 0$: equal endpoint values, and f is continuous and differentiable everywhere, so Rolle's Theorem applies. $f'(x) = 2x - 4 = 0$ gives $c = 2$, which lies in $(1, 3)$.
- 5 Explain why Rolle's Theorem does NOT apply to $f(x) = |x - 2|$ on $[0, 4]$, even though $f(0) = f(4) = 2$.
 f is not differentiable at $x = 2$ (it has a corner), so one of Rolle's Theorem's hypotheses (differentiability on the open interval) fails — the theorem's guarantee does not apply.

The Mean Value Theorem

- 6 For $f(x) = x^3$ on $[0, 2]$, find the value(s) of c guaranteed by the Mean Value Theorem.
Average rate of change: $\frac{f(2) - f(0)}{2 - 0} = \frac{8}{2} = 4$. Need $f'(c) = 3c^2 = 4$: $c^2 = \frac{4}{3}$, $c = \frac{2}{\sqrt{3}} \approx 1.15$ (taking the positive root within $[0, 2]$).

- 7 A car travels 180 km in 2 hours. Using the Mean Value Theorem, explain why the car must have been traveling at exactly 90 km/h at some instant.
- Average velocity $= \frac{180}{2} = 90$ km/h. Since position is a differentiable function of time, the MVT guarantees some instant c where the instantaneous velocity $v(c) = 90$ km/h exactly.
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Using the MVT to bound function values

- 8 Given $f(2) = 5$ and $-3 \leq f'(x) \leq 4$ for all x , use the MVT to find the range of possible values for $f(6)$.
- By the MVT, $f(6) - f(2) = f'(c)(6 - 2) = 4f'(c)$ for some $c \in (2, 6)$. Since $-3 \leq f'(c) \leq 4$:
 $-12 \leq f(6) - 5 \leq 16$, so $-7 \leq f(6) \leq 21$.
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Consequences of the MVT: monotonicity

- 9 Explain, using the MVT, why $f'(x) > 0$ on an interval implies f is strictly increasing on that interval.
- For any $a < b$ in the interval, the MVT gives $\frac{f(b) - f(a)}{b - a} = f'(c)$ for some $c \in (a, b)$. Since $f'(c) > 0$ and $b - a > 0$, we get $f(b) - f(a) > 0$, i.e. $f(b) > f(a)$ — so f is strictly increasing.
- 10 Explain why two functions f and g with $f'(x) = g'(x)$ for all x must differ by a constant, using the MVT.
- Let $h(x) = f(x) - g(x)$. Then $h'(x) = f'(x) - g'(x) = 0$ for all x . If h were not constant, the MVT applied between any two points where h differs would give $h'(c) \neq 0$ for some c , a contradiction. So $h(x) = C$ for some constant C , i.e. $f(x) = g(x) + C$.